Joshua M. Sims

Computational chemistry of catalysis at complex interfaces

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Computational chemist specialising in catalysis at complex interfaces: speciation and dynamics of supported metal species, solvation effects on reactivity, and free-energy methods including QM/MM, enhanced sampling and machine-learned interatomic potentials. Trained across periodic and molecular electronic structure, with recurring collaborations in which computation is embedded in experimental studies.

POSITIONS

Oct 2025 to present	**Postdoctoral researcher** KU Leuven, Belgium. Group of Prof. Jeremy Harvey, quantum and physical chemistry division. Mechanisms and kinetics in computational catalysis. Computational work within the iBOF consortium (De Feyter, Maes, Dusselier, Harvey) on deoxydehydration and supported rhenium catalysts.
Sep 2022 to Aug 2025	**Doctoral researcher** ENS de Lyon, France. Group of Dr Carine Michel. Speciation and morphology of Ru clusters on TiO_2, metal-support interactions, solvation at oxide interfaces, free-energy methods.
Nov 2021 to Jul 2022	**Research-internship year** Durham University, United Kingdom, then Université Grenoble Alpes, France. A full year of external research placements required by the ENS curriculum. With Dr Basile Curchod, the limits of TD-DFT for atmospheric photochemistry (Newton-X, Turbomole). With Prof. Anne Milet, computational study of CO_2-reduction catalysts on graphite (CP2K, DockOnSurf).
2019 to 2021	**Earlier research placements** ENS de Lyon, and Université de Rennes 1, France. With Dr Stephan Steinmann, resampling techniques for adsorption free energies (SolvHybrid, AMBER, VASP). With Dr Carine Michel, alcohol adsorption on alumina by metadynamics (CP2K, PLUMED) and water-induced deactivation of metathesis catalysts. With Prof. Florence Mongin and Dr William Erb, an experimental placement on the enantioselective synthesis of ferrocenes.

EDUCATION

2022 to 2025	**PhD in Chemistry, ENS de Lyon** Supervisor Dr Carine Michel. Thesis: morphology and surface state of a Ru/TiO_2 catalyst, a computational approach. Defended unconditionally on 17 July 2025. Funded by a competitive CDSN doctoral scholarship awarded on national competition.
2019 to 2021	**MSc in Computational Chemistry, ENS de Lyon**
2018 to 2019	**Diploma in Chemistry and Physics, ENS de Lyon** Highest honours, mention très bien. Admitted through the national competitive entrance examination, with salaried student status as a normalien.
2016 to 2018	**Classes préparatoires, Lycée Chateaubriand, Rennes** Intensive university-level programme in mathematics, physics and chemistry. Ranked first in class.

PEER-REVIEWED PUBLICATIONS

1. **Sims, J. M.**; Michel, C. Influence of oxygen vacancies on the morphology of Ru_{10} clusters supported on TiO_2. *Phys. Chem. Chem. Phys.* **2025**, 27, 17564-17571. 10.1039/D5CP01579K
2. Utama, J. K.; **Sims, J. M.**; Michel, C.; Giorgi, J. B. Hydration of hexafluoroacetone on ice. *J. Phys. Chem. C* **2025**, 129, 5868-5875. 10.1021/acs.jpcc.5c00560
3. Berg, I.; Mondal, R.; **Sims, J. M.**; Ben-Tzvi, T.; Lahav, L.; Friedman, B.; Michel, C.; Nairoukh, Z.; Gross, E. Strong substrate-adsorbate interactions direct the impact of fluorinated N-heterocyclic carbene monolayers on Au surface properties. *ACS Appl. Mater. Interfaces* **2024**, 16, 65469-65479. 10.1021/acsami.4c12514
4. Arjunan, S.; **Sims, J. M.**; Duboc, C.; Maldivi, P.; Milet, A. Investigating the interplay between charge transfer and CO_2 insertion in the adsorption of a NiFe catalyst for CO_2 electroreduction on a graphite support. *J. Comput. Chem.* **2024**, 45, 1690-1696. 10.1002/jcc.27355
5. Blanco, C. O.; **Sims, J.**; Nascimento, D. L.; Goudreault, A. Y.; Steinmann, S. N.; Michel, C.; Fogg, D. E. The impact of water on Ru-catalyzed olefin metathesis: potent deactivating effects even at low water concentrations. *ACS Catal.* **2021**, 11, 893-899. 10.1021/acscatal.0c04279

6. Lassagne, F.; **Sims, J. M.**; Erb, W.; Mongin, O.; Richy, N.; El Osmani, N.; Fajloun, Z.; Picot, L.; Thiéry, V.; Robert, T.; Bach, S.; Dorcet, V.; Roisnel, T.; Mongin, F. Thiazolo[5,4-f]quinoxalines, oxazolo[5,4-f]quinoxalines and pyrazino[b,e]isatins: synthesis from 6-aminoquinoxalines and properties. *Eur. J. Org. Chem.* **2021**, 2756-2763. `10.1002/ejoc.202100362`

FUNDING AND COMPUTING ALLOCATIONS

- **VSC Tier-1, Flanders, 2026.** Starter project on Hortense, 500,000 core-hours, and a pilot project on Sofia, the new Flemish national supercomputer. Both on peer-reviewed proposal.
- **GENCI, France, 2022 to 2025.** Three allocations on the French national supercomputers, 500,000 CPU-hours each, 1.5 million CPU-hours in total.
- **CDSN doctoral scholarship, 2022 to 2025.** Contrat Doctoral Spécifique Normalien, awarded by the ENS de Lyon on national competition.
- **Competitive examinations.** Admitted to the ENS de Lyon, ENS Paris-Saclay and ESPCI by concours, carrying salaried student status. Second in the French national selection for the International Chemistry Olympiad, 2017.

TEACHING, SUPERVISION AND MENTORING

- **Teaching assistant, ENS de Lyon, 2022 to 2025.** 64 hours per year: statistical-physics tutorials (third-year undergraduate, 26 h); periodic-DFT practicals (first-year master's, 12 h); organisation and supervision of the computational-chemistry practicals (second-year master's, 26 h), including direct supervision of student project work and the setting and marking of theoretical-chemistry examinations. Tutoring for the agrégation preparation. Teaching in French and English.

CONFERENCE CONTRIBUTIONS

- **Oral.** International Conference on Theoretical Aspects of Catalysis (ICTAC), Seville, 2024. Morphology of Ru nanoparticles at the titania-water interface.
- **Oral.** Rencontres Théomsia-RCTF, the French computational-chemistry network, 2024. Same topic.
- **Poster.** 18th International Congress on Catalysis (ICC), Lyon, 2024. Same topic.
- **Poster, co-author.** 16th European Inorganic Chemistry Conference (EICC), 2022. CO_2 reduction to CH_4 by a bio-inspired NiFe hydrogenase model on graphite.

PEER REVIEW, OUTREACH AND SERVICE

- Reviewer for the *Journal of the American Chemical Society*.
- Co-author of a problem for the French national selection for the 52nd International Chemistry Olympiad, 2020. Volunteer scientific administrator, 51st International Chemistry Olympiad, Paris, 2019.

TECHNICAL SKILLS AND LANGUAGES

Electronic structure	VASP, CP2K, Gaussian, ORCA, Turbomole, PySCF
Free energy	AMBER, PLUMED (metadynamics, OPES), QM/MM free-energy perturbation, resampling estimators
Learned potentials	Fine-tuning of foundation models (MACE, UMA), ANI
Computing	Python (NumPy, the ASE ecosystem, JAX), workflow automation (DockOnSurf, SolvHybrid), HPC on the GENCI national facilities and VSC Flanders
Languages	English (native speaker, Cambridge C2). French (bilingual). German (B1).
Nationality	French and British